

Hands-On vs. High-Tech:
How AI is Reshaping Physical Therapy Practices
Ben Dignan
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Ben Dignan
Mrs. Toni Ireland
Intern/Mentor
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Abstract

Artificial intelligence is becoming increasingly integrated into healthcare systems, including physical therapy and rehabilitation sciences. This paper investigates the role of AI in physical therapy, focusing on both the benefits and limitations of emerging technologies such as machine learning movement analysis, AI-assisted documentation systems, predictive recovery models, and tele-rehabilitation platforms. Secondary research from scholarly journals, medical publications, and healthcare technology sources was synthesized alongside primary research collected through interviews with professionals and students involved in physical therapy and rehabilitation. Interview participants included licensed physical therapists, professors, healthcare researchers, and graduate students. The findings reveal that AI has significant potential to improve efficiency, documentation, movement analysis, and patient accessibility. However, the interviews and literature consistently emphasized that artificial intelligence cannot replace essential human components of therapy such as clinical reasoning, empathy, ethical judgment, and manual treatment techniques. Concerns regarding bias, reliability, data privacy, and overreliance on automation were also frequently discussed. Overall, this research suggests that AI will likely function as a supportive clinical tool rather than a replacement for therapists. The future of physical therapy will likely involve collaboration between human clinicians and intelligent systems while maintaining patient-centered care.

Literature Review

Last year, Bill Gates made waves in the clinical world when he appeared on "The Tonight Show", claiming "within 10 years, AI will replace many doctors and teachers—humans won't be needed 'for most things'". AI models have proven to be extremely beneficial in taking notes, handling medical records, and predicting patient outcomes. Models including Noterro have been adopted for tracking medical records. Several tools and apps like Kaia Health and Physitrack have been developed further by AI to guide patients in exercises virtually through their mobile device. On the other hand, many professionals have concerns over ethics and biases of AI, and patients' lack of trust may hinder the implementation of AI. While the mystery of AI's capability to assist or threaten the roles of human therapists remains ongoing, this paper examines all the factors involved. Although artificial intelligence is being increasingly used in physical therapy for movement analysis and increasing efficiency, it cannot replace clinical judgment or ethical decision making, making human therapist oversight essential for safe and effective patient rehabilitation.

AI's rise in physical therapy began gradually, with the first applications of AI being administrative functions like recordkeeping and documentation. Since then, AI's abilities have grown to encompass movement analysis, rehabilitation assistance, and recovery prediction. These advances hold promise for patients and therapists alike. Benefits include: increased efficiency, broader access to care, and better regulation with exercise programs. For example, pose-estimation models can accurately track joint positions and balance, and robotic rehabilitation systems have achieved an accuracy score of 0.81 in post-stroke exercise monitoring (Roggio, 2024). However, while the application of AI technology across various industries must be evaluated and its accuracy needs to be tested, there are additional concerns surrounding bias and loss of human judgment in clinical decision-making.

The effectiveness of AI can heavily depend on the context. Many advanced real-time tracking systems struggle with complex movements, and most clinics lack the technology to implement these solutions effectively (Kapica, 2024). AI can spot mistakes like bad form, but it generally struggles with more complex corrections that require knowledge and empathy for patient pain. In the same way, AI can flag incorrect squat form but does not adjust instructions based on a patient's fatigue or anxiety (Alshami et al., 2025). Still, AI-guided exercise has been shown to reduce pain by 30% more than standard care alone (Areias et al., 2024). In addition, AI contributes to efficiency and clinical optimization by reducing repetitive tasks. Automated documentation tools allow therapists to dedicate more time to patient interaction by diminishing note-taking time by 50–70% (Net Health, 2024). Overall, machine learning exercise programs have proven to strengthen patient results, such as more pain reduction compared with standard care when the exercise is implemented in conjunction with therapist oversight rather than in replacement, embodying how AI can complement rehabilitation but is not a valid substitute for the professional clinicians' judgment and individualized approach.

Artificial intelligence is changing the professional role of physical therapists greatly in how clinical decisions are made and how therapists interact with technology in their daily work. While AI systems can assist in analyzing data and suggesting treatment patterns, most experts emphasize that AI should function as a clinical assistant. Therapists hold legal and ethical accountability for treatment decisions, and clinicians must always interpret and validate AI-generated outputs (AI in Therapy, 2025). Even when AI systems offer valuable perspectives on movement dynamics or rehabilitation advancement, therapists are obligated to carefully assess recommendations to ensure their safety for each specific patient. Within healthcare environments, the practitioner bears the ultimate responsibility for patient outcomes, underscoring the crucial need for human supervision when incorporating AI into treatment strategies (Aliferis & Simon, 2024).

Beyond legal obligations, numerous therapists consider the preservation of authority as important for the profession. Surveys investigating physical therapists' perspectives on artificial intelligence have revealed that the majority of clinicians prefer to maintain ultimate authority over diagnoses and treatment strategies (Physical Therapists' Attitudes Toward Artificial Intelligence Diagnostics, 2025). Therapists frequently voice concerns that an over-reliance on automated systems could erode clinical reasoning abilities. If AI assumes a primary role in patient data analysis, there is potential for therapists to become overly dependent on technology, hindering the development of their own clinical judgment through practical experience. Furthermore, research into therapists' perceptions of artificial intelligence within healthcare has indicated that certain clinicians harbor concerns regarding the long-term consequences of AI on professional autonomy and expertise (Physical Therapists' Perceptions and Attitudes Toward Artificial Intelligence in Healthcare and Rehabilitation, 2025). These concerns underscore the necessity of technological assistance and the continued development of human clinical skills.

Instead of eliminating the need for therapists, AI appears to be reshaping how therapists spend their time and how they deliver care. In many cases, AI technologies take over repetitive administrative tasks such as documentation, recordkeeping, and data organization. This shift allows therapists to focus more on patient interaction, manual treatment techniques, and complex clinical reasoning. According to research on the use of artificial intelligence in physical therapy, clinicians increasingly view AI as a supportive tool that enhances their efficiency rather than replacing their role in patient care (University of St. Augustine for Health Sciences, 2021). Consequently, the therapist's responsibilities could shift to encompass the oversight of AI-generated data, the interpretation of its findings, and the formulation of ultimate treatment strategies, integrating technological insights with professional expertise.

Furthermore, AI's influence on physical therapy extends to enhancements in both operational efficiency and patient accessibility. A primary advantage of AI technology is its

capacity to substantially alleviate the administrative workload shouldered by clinicians.

Documentation, a particularly time-intensive facet of clinical practice, frequently necessitates that therapists dedicate considerable time to note-taking and the updating of patient records following each session. AI-powered documentation tools can automate much of this process by generating clinical notes based on recorded conversations or structured inputs. Studies examining AI documentation systems suggest that these tools can reduce note-writing time by approximately 50–70 percent, allowing therapists to dedicate more time to direct patient care (Net Health, 2024). The streamlining of administrative tasks has the potential to enhance clinic productivity, thereby enabling therapists to concentrate on rehabilitation interventions that directly impact patient outcomes.

Additionally, artificial intelligence can broaden the availability of physical therapy, especially for individuals unable to attend regular in-person sessions. Remote rehabilitation platforms facilitate home-based exercise regimens, offering feedback via smartphone cameras, wearable sensors, or motion tracking technologies. These systems are designed to monitor patient movements, offering corrective guidance when exercises are executed improperly. In certain instances, AI-driven systems can even identify performance trends and suggest modifications to exercise protocols. Research into AI-enhanced rehabilitation tools has demonstrated that wearable devices and remote monitoring platforms can facilitate the extension of therapeutic interventions beyond traditional clinical settings, enabling therapists to remotely oversee patient advancement (Silva & Costa, 2023). This functionality holds particular significance for patients residing in rural locales or those with mobility constraints, who may encounter difficulties in accessing therapy clinics.

However, the efficiency improvements afforded by AI are not universally experienced across all healthcare environments. Certain clinics may be deficient in the technological infrastructure necessary for the deployment of sophisticated AI systems, encompassing high-quality sensors, motion tracking apparatus, or specialized software. Smaller practices or clinics with limited financial resources may encounter challenges in acquiring these technologies, potentially leading to disparities in access to AI-assisted rehabilitation tools. In addition, some AI tracking systems still struggle with highly complex or unpredictable movements, meaning that they cannot fully replace in-person observation by trained therapists. These limitations suggest that while AI can significantly improve efficiency and access to care, it should be viewed as a complement to traditional therapy rather than a universal solution for all clinical environments (Kapica, 2024).

While the existing literature provides substantial information regarding AI systems and rehabilitation technologies, further primary research was necessary to better understand how professionals currently working in the field perceive the rapid integration of AI into physical therapy practice. Scholarly literature often focuses heavily on technical capabilities and clinical outcomes, but less attention is given to how therapists, professors, researchers, and students personally view AI's role in rehabilitation settings. In order to bridge this gap, interviews were conducted with individuals involved in physical therapy, rehabilitation science, sports medicine, and healthcare research. These interviews were intended to provide insight into how professionals interpret the ethical concerns, clinical limitations, and practical applications of artificial intelligence within real-world healthcare environments.

Methods

Primary research for this study was conducted through qualitative interviews with five individuals involved in physical therapy, rehabilitation, sports medicine, and healthcare research. Participants were selected based on their professional or academic involvement with rehabilitation sciences, physical therapy education, or AI technology. The sample included licensed physical therapists, a graduate student pursuing a Doctor of Physical Therapy degree, a university professor, a chiropractic and sports medicine specialist, and a researcher familiar with artificial intelligence systems.

The interviews were conducted virtually due to scheduling and geographic limitations. Participants were asked a series of open-ended questions regarding artificial intelligence in physical therapy. Topics included movement analysis, AI-assisted documentation systems, electronic medical records, SOAP notes, predictive recovery models, clinical reasoning, bias in machine learning systems, accessibility of care, and the future role of AI in rehabilitation settings. Questions were slightly adjusted depending on each participant's professional background, and their responses were transcribed, summarized, and analyzed for patterns, similarities, and differences among participants.

Results

The interviews revealed several consistent themes regarding the role of artificial intelligence in physical therapy and rehabilitation. Nearly every participant believed that AI would become increasingly useful in administrative tasks such as documentation, SOAP notes, billing, and electronic medical records. Dr. Plotkin explained that AI is already helping healthcare professionals with typing reports, letters of medical necessity, treatment notes, and paperwork. Dr.

Ford similarly stated that AI could save therapists time by assisting with documentation and allowing clinicians to spend more time interacting with patients.

Participants also consistently emphasized that artificial intelligence cannot replace the human aspects of physical therapy. Dr. Ford explained that her work relies heavily on manual therapy techniques and direct physical interaction with patients, which she believes AI cannot replicate. Ms. DeSola discussed the importance of empathy, communication, and emotional awareness when working with patients experiencing difficult medical or financial situations. Multiple participants stated that clinical reasoning, ethical judgment, and patient communication remain uniquely human skills.

Several interviewees discussed the growing role of AI in movement analysis and predictive rehabilitation systems. Mr. Mariano described how AI can analyze motion data from wearables, motion-capture systems, and smartphone sensors to identify patterns related to pain, fatigue, posture, and recovery progress. Dr. Khan similarly explained that AI shows promise in outcome prediction, tele-rehabilitation, and movement analysis integrated with motion analysis laboratories.

At the same time, concerns regarding data collection and reliability were repeatedly identified. Dr. Khan explained that rehabilitation datasets are difficult to maintain because rehabilitation outcomes are highly individualized and involve extensive condition-specific measurements. Mr. Mariano also emphasized that patient data can vary significantly depending on age, injury type, motivation, disability, and access to healthcare. Several participants raised concerns regarding bias in AI systems if training data underrepresents certain patient populations.

Another major trend involved the belief that AI should function as an assistant rather than a replacement for therapists. Participants consistently stated that AI can improve efficiency and assist with analysis but should remain under human supervision. Multiple interviewees emphasized that therapists must continue validating AI-generated outputs and maintaining responsibility for patient outcomes.

Discussion

The findings from both the literature review and the interviews strongly support the thesis that artificial intelligence can improve physical therapy practices but cannot replace human therapists. Across both secondary and primary research, the most common theme was that AI functions best as a supportive clinical tool rather than an independent decision-maker.

The interviews reinforced many of the concerns discussed in the scholarly literature. Researchers such as Dr. Khan and Mr. Mariano focused heavily on AI's ability to analyze large datasets, predict outcomes, and improve rehabilitation monitoring through motion analysis and wearable technologies. Their perspectives aligned closely with the academic literature discussing machine learning models, predictive recovery systems, and remote rehabilitation technologies. At the same time, practicing clinicians such as Dr. Ford and Dr. Plotkin emphasized the practical realities of patient care, particularly the importance of manual therapy, clinical reasoning, and direct human interaction.

Although participants came from different professional backgrounds, several consistent themes emerged. Researchers focused more heavily on predictive analytics, motion analysis, and tele-rehabilitation, while practicing clinicians emphasized documentation efficiency and the

irreplaceable value of hands-on care. Despite these differences, all participants agreed that AI should function as a supportive tool rather than an independent replacement for therapists.

Another major finding involved the growing importance of AI-assisted documentation systems. Nearly all participants acknowledged that documentation remains one of the most time-consuming aspects of clinical practice. The literature review demonstrated that AI systems can significantly reduce documentation time, and the interviews further supported this claim. Participants repeatedly stated that reducing administrative workload could allow therapists to dedicate more time to patient interaction and rehabilitation.

The interviews also highlighted concerns regarding reliability, bias, and ethical responsibility. Multiple participants warned that AI systems remain heavily dependent on the quality and diversity of their training data. If datasets fail to adequately represent different patient populations, AI-generated recommendations could become inaccurate or inequitable. These concerns closely reflect existing literature discussing bias in healthcare algorithms and the ethical implications of machine learning systems.

This research also revealed several limitations. First, the sample size was relatively small, consisting of only five interview participants. In addition, participants represented a limited range of specialties within rehabilitation sciences and healthcare. Some participants had extensive familiarity with AI systems, while others had limited direct experience using AI in practice. Furthermore, because interviews were conducted through text and email communication, responses were generally shorter than what might have been obtained through in-person interviews.

Despite these limitations, the interviews provided valuable insight into how professionals currently perceive artificial intelligence within physical therapy. Future research could expand upon this study by conducting larger interviews across multiple healthcare disciplines or by surveying patients regarding their comfort level with AI-assisted rehabilitation systems. Additional studies may also investigate how AI technologies impact patient outcomes over long periods of rehabilitation.

Overall, the evidence suggests that the future of physical therapy will likely involve collaboration between human clinicians and intelligent systems. Artificial intelligence may continue improving movement analysis, remote rehabilitation, documentation systems, and predictive recovery models. However, therapists will continue to play an essential role in ethical decision-making, clinical reasoning, manual treatment, and emotional support.

Conclusion

Artificial intelligence is rapidly transforming many aspects of physical therapy. Machine learning technologies can analyze movement patterns, automate documentation, and assist therapists in managing large amounts of patient data. These tools have the potential to improve efficiency within clinics and expand access to rehabilitation services through remote monitoring and guided exercise programs.

Despite these advantages, the evidence reviewed in this paper suggests that artificial intelligence cannot replace the essential human elements of physical therapy practice. Manual therapy techniques, clinical reasoning, and empathetic communication remain pillars of effective

patient care. Interviews with professionals in the field reinforce this conclusion, highlighting that therapists view AI as a helpful support tool rather than a replacement for their expertise.

As AI technology continues to evolve, the future of physical therapy will likely involve collaboration between human clinicians and intelligent systems. Therapists will continue to provide their own human aptitudes, such as clinical judgment, manual treatment, and emotional support, while artificial intelligence assists with data analysis and administrative tasks. The challenge facing physical therapy will not be whether AI can replace therapists, but how the profession can responsibly integrate technology while preserving the human connection that rehabilitation depends upon.

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Appendices

Appendix A: Sample Interview Questions

What are the most promising areas of research involving AI in physical therapy or rehabilitation science today?

How has AI changed the way researchers analyze human movement and rehabilitation outcomes?

Many AI systems rely on large datasets of patient movement and recovery outcomes. What challenges exist when collecting or training AI models using rehabilitation data?

How reliable are current AI-based movement analysis tools compared to traditional assessment by trained therapists?

What limitations currently prevent AI from replacing clinical reasoning performed by physical therapists?

How has AI improved documentation systems such as SOAP notes or electronic medical records in physical therapy practice?

Some researchers have raised concerns about bias in medical AI systems. How might bias affect AI tools used in rehabilitation settings?

What role should physical therapists play in guiding the ethical use of AI technologies in healthcare?

How do you personally envision AI influencing physical therapy over the next decade?

Appendix B: Interview Participants

Dr. Fayaz Khan – Professor and rehabilitation researcher

Marciano Mariano – Healthcare researcher and AI policy advisor

Dr. Larry Plotkin – Chiropractor and sports medicine specialist

Dr. Tami Ford – Licensed physical therapist

Dani DeSola – Doctor of Physical Therapy graduate student